

Module 1

DC Circuit Analysis and Theorems

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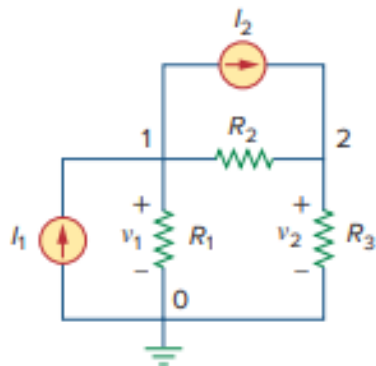
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Nodal analysis

In nodal analysis, we are interested in finding the node voltages.

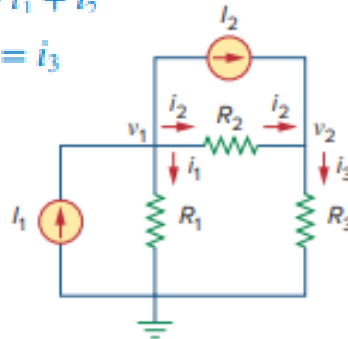
Steps to determine Node Voltages:

1. Select a node as the reference node.
2. Assign voltages $v_1, v_2, \dots, v_n - 1$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
3. Apply KCL to each of the $n - 1$ not referenced nodes. Use Ohm's law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.



$$I_1 = I_2 + i_1 + i_3$$

$$I_2 + i_2 = i_3$$



$$i_1 = \frac{v_1 - 0}{R_1} \quad \text{or} \quad i_1 = G_1 v_1$$

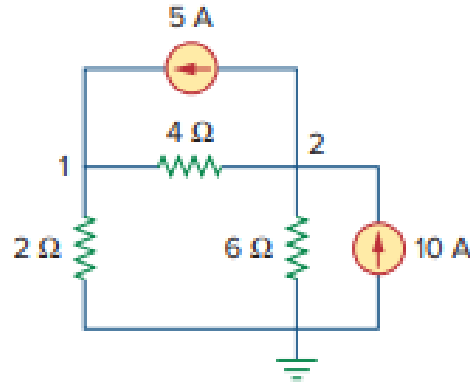
$$i_2 = \frac{v_1 - v_2}{R_2} \quad \text{or} \quad i_2 = G_2 (v_1 - v_2)$$

$$i_3 = \frac{v_2 - 0}{R_3} \quad \text{or} \quad i_3 = G_3 v_2$$

$$\begin{aligned} I_1 &= I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2} \\ I_2 + \frac{v_1 - v_2}{R_2} &= \frac{v_2}{R_3} \end{aligned} \quad \left| \quad \begin{aligned} I_1 &= I_2 + G_1 v_1 + G_2 (v_1 - v_2) \\ I_2 + G_2 (v_1 - v_2) &= G_3 v_2 \end{aligned} \right.$$

Nodal Analysis

Calculate the node voltages in the circuit



At node 1,

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

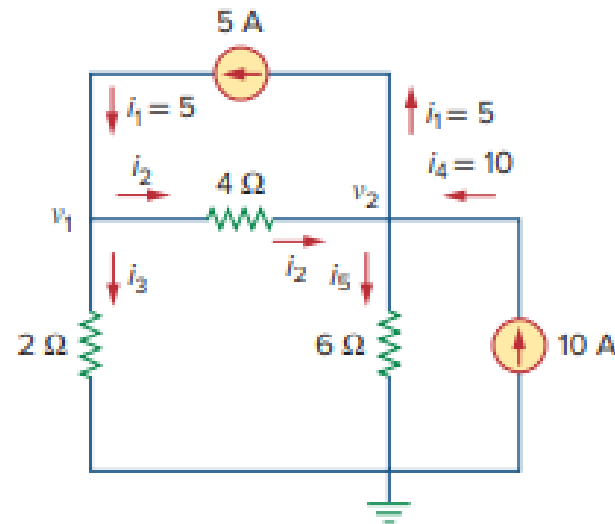
$$20 = v_1 - v_2 + 2v_1$$

At node 2,

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

$$-3v_1 + 5v_2 = 60 \quad \text{-----2}$$



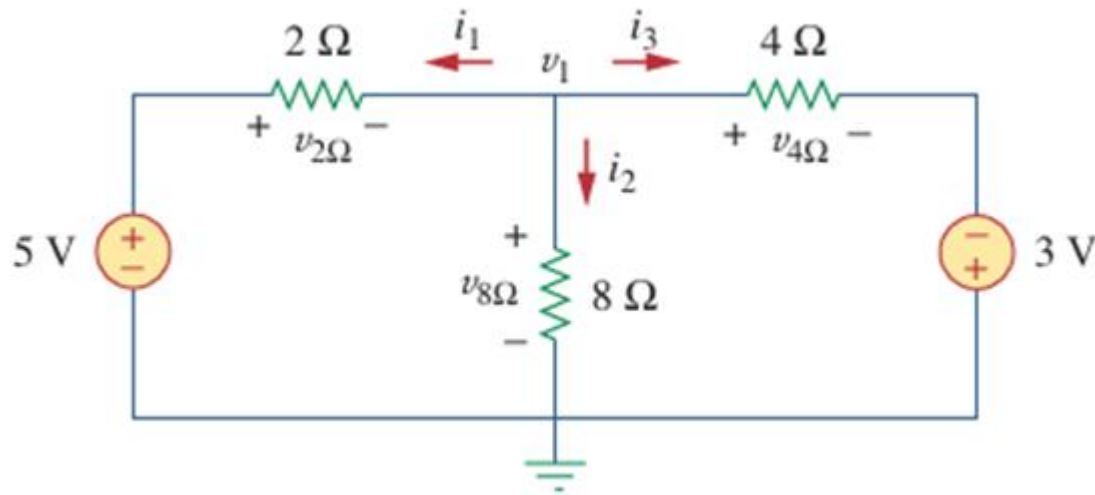
Solve the 2 simultaneous equations:

$$v_1 = 13.33 \text{ V}$$

$$v_2 = 20 \text{ V}$$

Nodal Analysis

Solve for $i_{8\Omega}$ using Nodal analysis method.



$$i_{8\Omega} = i_2, \quad i_2 = \frac{v_1}{8}, \quad i_{8\Omega} = \frac{v_1}{8}$$

$$\frac{v_1 - 5}{2} + \frac{v_1 - 0}{8} + \frac{v_1 + 3}{4} = 0$$

Now we can solve for v_1 .

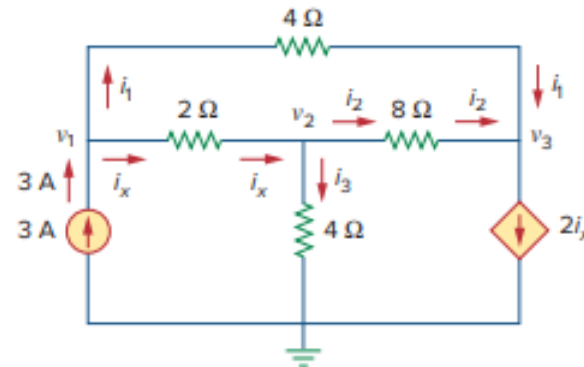
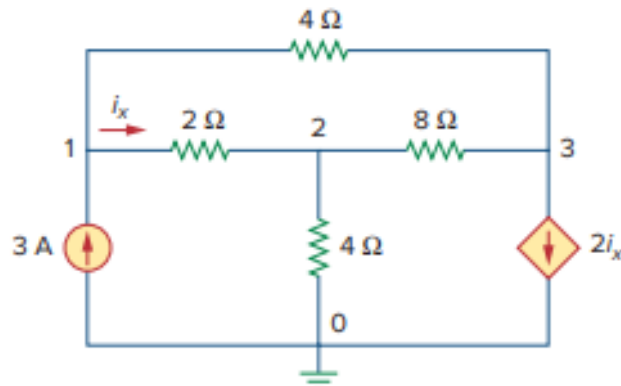
$$8 \left[\frac{v_1 - 5}{2} + \frac{v_1 - 0}{8} + \frac{v_1 + 3}{4} \right] = 0$$

$$\text{leads to } (4v_1 - 20) + (v_1) + (2v_1 + 6) = 0$$

$$7v_1 = +14, \quad v_1 = +2 \text{ V}, \quad i_{8\Omega} = \frac{v_1}{8} = \frac{2}{8} = \mathbf{0.25 \text{ A}}$$

Nodal Analysis

Determine the voltages at the nodes in the circuit.



At node 1, $3 = i_1 + i_x \Rightarrow 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$

$3v_1 - 2v_2 - v_3 = 12$ -----1

At node 2, $i_x = i_2 + i_3 \Rightarrow \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$

$-4v_1 + 7v_2 - v_3 = 0$ -----2

At node 3, $i_1 + i_2 = 2i_x \Rightarrow \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$

$2v_1 - 3v_2 + v_3 = 0$ -----3

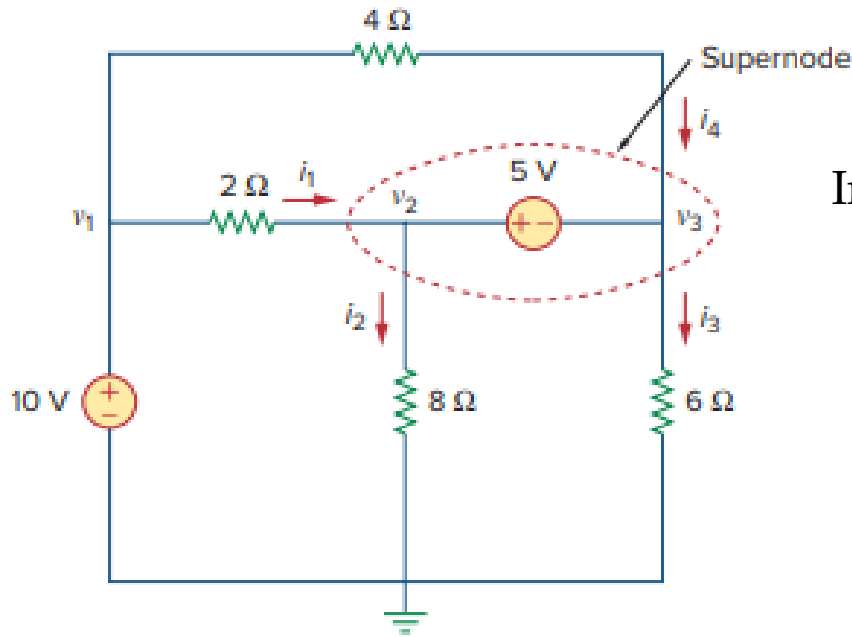
Solve the 3 simultaneous equations:

$v_1 = 4.8 \text{ V}$

$v_2 = 2.4 \text{ V}$

$v_3 = -2.4 \text{ V}$

Super Node



In this circuit nodes 2 and 3 form a supernode.

Case 1: A voltage source is connected between the reference node and a non reference node.

$$V_1 = 10 \text{ V} \quad \text{-----} 1$$

Case 2: A voltage source (dependent or independent) is connected between two non reference nodes, the two non reference nodes form a generalized node or supernode; we apply both KCL and KVL to determine the node voltages.

Super Node

An essential component of nodal analysis is applying KCL, which requires knowing the current through each element.

There is no way of knowing the current through a voltage source in advance.

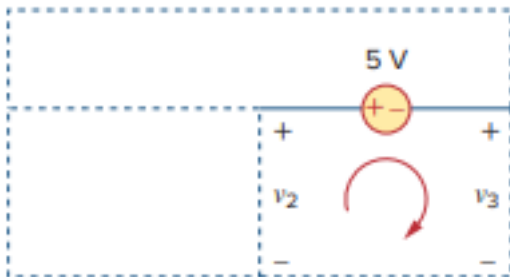
However, KCL must be satisfied at a supernode like any other node.

Hence, at the supernode, $i_1 + i_4 = i_2 + i_3$

or

$$\frac{v_1 - v_2}{2} + \frac{v_1 - v_3}{4} = \frac{v_2 - 0}{8} + \frac{v_3 - 0}{6} \quad \text{-----2}$$

To apply KVL to the supernode, redraw the circuit as shown and write the equation using KVL.



$$-v_2 + 5 + v_3 = 0 \quad \Rightarrow \quad v_2 - v_3 = 5 \quad \text{-----3}$$

Solving these 3 equations, you can find the node voltages.

Next class: Mesh Analysis